CHAPTER 5

Non-Infectious Diseases in Asian Elephants

Toxicology, Electrocution, Nutritional Disorders & Field Investigation Framework

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1. Background — Why Elephants Are Biologically Distinct

Asian elephants are physiologically unique in two seemingly contradictory respects: they are extraordinarily resistant to cancer, yet highly vulnerable to poisoning. Understanding both traits is essential for interpreting necropsy findings and identifying non-infectious causes of mortality.

Cancer Resistance — Peto's Paradox

Despite enormous body size and a 60–70 year lifespan (far more cells and far more time for mutations than in small animals), elephants have a **remarkably low incidence of cancer**. The key mechanism:

- **Multiple TP53 copies** — elephants carry **20+ copies** of the TP53 tumour-suppressor gene ("guardian of the genome"), compared with just **1 copy in humans**.
- **Enhanced DNA-damage detection** — more TP53 enables faster, more accurate identification of cellular DNA errors.
- **Stronger apoptotic response** — damaged cells are eliminated by programmed self-destruction before they can proliferate into tumours.

High Susceptibility to Toxicity

The same large body that should theoretically raise cancer risk instead makes elephants **highly prone to poisoning**. Several converging factors explain this:

Risk Factor	Mechanism / Consequence
Dysfunctional CYP genes	Cytochrome P450 (CYP) genes are dysfunctional → limited ability to biotransform and detoxify ingested toxins
Massive daily intake	Adults consume 100–300 kg of vegetation per day; toxic plants, contaminated fodder, pesticides, and mycotoxins are ingested in large amounts
High water consumption	100–200 L of water per day — polluted sources expose them to heavy metals, pesticides, cyanobacterial and industrial toxins
Hindgut fermenters	Sudden diet changes or spoiled feed disrupt gut microbiota and increase endogenous toxin production
Non-selective feeding	Explore environment with the trunk; may consume unfamiliar plants, crops, garbage, or treated material without discrimination
Drug sensitivity	Extremely sensitive to certain veterinary drugs and anaesthetics; incorrect dosing can cause respiratory depression, cardiac failure, or death
Pollutant accumulation	Long lifespan (60–70 yrs) allows progressive build-up of heavy metals (lead, mercury, arsenic) and persistent organic pollutants

2. Case Study — Mass Mortality at Bandhavgarh NP (Oct 2024)

History & Field Observations

Incident: Sudden death of **10 elephants from a herd of 13** on **31 October 2024**. During forest-department patrolling, several elephants were found recumbent with weak respiration; some were already dead.

Location: Salkhania beats of Pataur and Khitauli Ranges, Bandhavgarh National Park (Tiger Reserve), Madhya Pradesh.

- **Heavy rainfall** during October 2024.
- **Kodo millet crops were unharvested** near forest boundaries — a critical epidemiological clue linking the elephants to fungal-contaminated grain.

Clinical Signs & Gross Necropsy

Clinical progression: respiratory distress → ataxia → weakness → incoordination → recumbency → rapid collapse.

Organ / Site	Gross Finding
Lungs	Congestion and oedema
Liver	Enlarged, congested and swollen
Kidneys	Congested
Body cavities	Serosanguinous (blood-tinged) fluid accumulation
Multiple organs	Congestion and haemorrhages throughout
Heart	Pale necrotic foci on the myocardium

Histopathology

Tissue	Histopathological Finding
Heart	Acute myocardiopathy — degeneration, vacuolation and necrosis of cardiac myofibres
Liver	Hepatocellular degeneration — centrilobular necrosis, fatty change
Kidney	Renal tubular necrosis with epithelial degeneration
General	Congestion, haemorrhage and oedema throughout multiple organs

Sample Collection & Investigation Pathway

Sample	Preservation	Purpose
Tissues in 10% NBF	Formalin fixation	Histopathology
Tissues on ice + heart blood	Chilled	Disease screening (EEHV, EMCV, HS, Clostridium)
Stomach contents, liver, kidney, heart, fat	—	Toxicology
Hair follicles / nails	—	Toxicology (chronic exposure indicators)
Environmental samples (water, feed, fodder, grain)	—	Toxicology

Toxicological methods: conventional tests → thin-layer chromatography (TLC) → **liquid chromatography–mass spectrometry (LC-MS)**.

Toxicology Results

Negative for: HCN, nitrate-nitrite, heavy metals, organophosphates, organochlorines, pyrethroids, carbamates.

Negative for infectious causes: EEHV, pasteurellosis, other mycotoxins.

Positive — Cyclopiazonic acid (CPA): detected in all pooled samples at **>100 ppb** (Ref. CAD-877/2024-25, dated 05-11-2024). Results indicated the elephants had consumed large quantities of Kodo plant/grains.

LC-MS Confirmation (ICRISAT, Hyderabad)

23 samples received 6 November 2024. High CPA detected in stomach contents, tissues, and millet samples:

Sample Description	CPA (ppb)
Kodo grain with straw (range)	~1,567 – 12,874
K-2 (Kodo grain + straw)	12,874
K-5 (Kodo grain + straw)	10,667
Vomit sample (C3, Kodo millet)	4,486

Final Diagnosis & Pathophysiology

DIAGNOSIS: Toxicosis — Kodo millet (cyclopiazonic acid) poisoning.
Pathophysiology: CPA caused acute cardiac injury (myocardiopathy) → cardiac arrest / heart failure → passive venous congestion → hypoxia → degeneration of parenchymatous organs (liver, kidney, spleen). Findings correlate fully with necropsy gross lesions and laboratory results.

3. Cyclopiazonic Acid (CPA) — Key Facts

Source, Chemistry & Toxicity

- **Source plant:** *Paspalum scrobiculatum* (kodo / koda millet). Grown mainly in Nepal; also India, Philippines, Indonesia, Vietnam, Thailand. Originated in West Africa.
- **CPA is a mycotoxin** — a secondary metabolite of fungi ***Aspergillus*** and ***Penicillium***. Kodo seed is particularly infected by *A. flavus* and *A. tamarii*.
- **Chemistry:** α-cyclopiazonic acid, an indole-tetramic acid; molar mass ≈ 336.39 g/mol; PubChem CID 54682463.
- **Key hazard:** Ripe kodo can become intermittently poisonous ("Kiruku Varagu") without any visible change — fitness for consumption cannot be judged by appearance alone.
- **First human report** linking CPA to kodua poisoning: Rao & Husain (1985), *Mycopathologia*.

Historical Precedent — 1933

R.C. Morris, *J. Bombay Nat. Hist. Soc.* (1934) documented the death of **14 elephants** near Vannathiparai Reserve Forest (December 1933) from kodo millet ("Varagu") poisoning. The traditional antidote was **tamarind-water or buttermilk in large quantities**; one of three elephants given water and tamarind reportedly survived (unconfirmed). This demonstrates that Kodo poisoning of elephants is a long-recognised, recurring problem.

Prevention & Control

1. **Regulate agriculture** and enhance monitoring of fungal-infected cropping systems near forest boundaries.
2. **Community awareness** — engage farmers and livestock owners around the reserve.
3. **Prevent elephant access** to contaminated, unharvested millet fields.
4. **Scientific surveillance** and systematic monitoring of kodo crops.
5. **Inter-departmental coordination** between forest, agriculture, and revenue departments.

Advisory & One Health Angle

Correlate history with necropsy; survey and destroy fungal-infected Kodo residue;
keep domestic animals and wildlife out of affected fields; conduct cropping/ambient studies;
determine the LD50 of CPA in domestic and wild animals.
ONE HEALTH ALERT: Cattle in the area were also affected. Watch for symptoms in humans —
Kodua poisoning is a recognised human mycotoxicosis.

4. Other Major Non-Infectious Causes of Elephant Mortality

A. Electrocution (Electrical Injury)

Electrocution is one of the most commonly reported non-infectious causes of elephant death in India, particularly from power lines and illegal electrified fences.

Mechanisms of damage:

- **Direct:** cell depolarisation, electroporation of cell membranes.
- **Indirect / thermal:** current generates heat → thermal burns and coagulative necrosis.
- **Secondary mechanical:** violent muscle contraction or cardiac arrest causes falls and crush injuries.
- Final common pathway: **cell death (necrosis) and organ failure**.

Comparison of Injury Patterns

Feature	Lightning	High Voltage (>1 kV)	Low Voltage (<1 kV)
Voltage	>30,000,000 V	>1,000 V	<1,000 V
Current	>200,000 A	<1,000 A	<240 A
Cardiac arrest	Asystole	Ventricular fibrillation	Ventricular fibrillation
Muscle contraction	Single (single jolt)	Depends on duration	Tetanic (sustained)
Burns	Rare, superficial	Common, deep	Usually superficial
Rhabdomyolysis	Uncommon	Very common	Common

Feature	Lightning	High Voltage (>1 kV)	Low Voltage (<1 kV)
Mortality	Very high	Moderate	Low

Current Magnitude & Physiological Effect

Reference thresholds (general): 3 mA — noticeable shock; 10 mA — muscular contractions; 30 mA — respiratory paralysis; 50 mA — heart paralysis (may be fatal); 100 mA — ventricular fibrillation (fatal); 4 A — heart paralysis (fatal); 5 A — tissue burning.

Tissue resistance determines the path and extent of injury. Nerves, blood, mucous membranes, and muscle have low resistance (higher injury). Tendons, fat, and bone have high resistance and are relatively spared.

Field Findings & Post-Mortem Clues

Of 10 reported electrocution cases in elephants: **6 from direct power-line contact (440–220 V)**, **4 from electrified fences**. Affected states: Uttar Pradesh (4), Tamil Nadu / Mudumalai NP (2), Uttarakhand (2), Chhattisgarh (2).

- **Very rapid decomposition** and quick or absent rigor mortis.
- **Lightning / burn marks** — charred skin, lacerations along the current path (e.g. trunk burns in the Dudhwa NP case).

B. Lightning Side-Potential (Herd Mortality)

A ground-strike lightning event creates a high-voltage point; current then spreads radially outward through the ground for **up to 100 m or more**. Elephants standing or lying on the ground become part of the electrical circuit. Current passes through the body between front and rear feet → **instant internal organ failure and death — without any direct lightning strike**. This mechanism explains multiple simultaneous deaths in closely grouped herds.

Field Clue — Lightning Side-Potential

High-risk conditions: open grasslands, proximity to water bodies, thunderstorms,
and tightly grouped herds.
Documented in Assam (pre-monsoon / monsoon season).
Investigate GPS clustering of carcasses and weather/storm records immediately.

C. Infighting / Tusk Injuries

Penetrating tusk wounds from intraspecific combat are a recognised non-infectious cause of elephant mortality, particularly in adult males. Post-mortem examination reveals deep puncture wounds, haemorrhage, and secondary infection. (e.g. Najibabad Forest Division, Uttar Pradesh.)

D. Nutritional Deficiencies

Nutritional deficiency disease can be **primary** (inadequate dietary supply) or **secondary**, caused by one of several interfering mechanisms:

Mechanism of Secondary Deficiency	Examples / Notes
Interference with intake	Pain on eating, dental disorders, jaw disease

Mechanism of Secondary Deficiency	Examples / Notes
Interference with absorption	Chronic intestinal disease, parasitism, dysbiosis
Interference with storage / utilization	Liver disease impairing vitamin storage and metabolism
Increased excretion	Renal disease, diarrhoea, heavy parasite burden
Increased requirements	Pregnancy, lactation, rapid growth in calves
Anti-nutritional substances	Dietary inhibitors (e.g. phytates, oxalates) blocking nutrient uptake

Clinical examples in elephants:

- **Hypocalcaemic tetany** and **rickets** / bone disorders (e.g. "Yogi" at Rajaji NP, Dehradun).
- **Corneal opacity and cataract** — chronic micronutrient deficiency.
- **Overgrown, cracked nails** and **cracked foot pads** — chronic nutritional and management deficiency in captive or managed animals.

E. Advanced Decomposition / Skeletonization

Late-discovered carcasses (e.g. Amangarh Tiger Reserve, Bijnor) substantially limit diagnostic yield. Advanced autolysis and scavenging destroy soft tissue lesions, preclude sampling for microbiology and virology, and compromise toxicological analysis. This underscores the critical importance of **early detection, rapid reporting, and prompt necropsy**.

5. Field Investigation Framework — Exam-Ready Checklist

When facing a suspected wildlife disease outbreak or mass mortality event, the wildlife veterinarian must be able to address these six areas:

Area	Key Actions / Considerations
1. First Response	Secure the site. Ensure personnel safety and PPE. Notify field director and authorities. Prevent further access/exposure. Barricade and photograph the scene.
2. Initial Field Investigation	Record complete history: time of onset, clinical signs observed, GPS coordinates, environmental context (recent rainfall, crops nearby, water sources). Note scavenger activity and any other species affected (One Health).
3. Essential Requirements	PPE and biosafety kit; necropsy and sampling kit (scalpels, forceps, containers); cold chain (ice box, gel packs); fixatives (10% NBF vials); documentation (camera, field book); labelled containers; transport logistics and lab contacts.
4. Samples for the Laboratory	Fixed tissues for histopathology; chilled tissues + heart blood for disease screening; stomach contents, liver, kidney, heart, fat, hair/nails for toxicology; environmental water, feed, fodder, grain.
5. Differential Diagnoses	Infectious: EEHV, Pasteurellosis/HS, EMCV, clostridial disease. Toxic: mycotoxins/CPA, HCN, nitrate-nitrite, pesticides, heavy metals. Physical: electrocution, lightning side-potential, tusk injuries. Nutritional/metabolic deficiency.
6. Prevention & Control	Apply the five-point plan (regulate agriculture, community awareness, prevent access, scientific surveillance, inter-departmental coordination). Operate under a One Health framework linking wildlife, livestock, and human health.

6. Key Takeaways

Chapter Summary — Non-Infectious Diseases in Asian Elephants

1. Elephants resist CANCER (20+ TP53 copies) but are highly VULNERABLE TO TOXINS (dysfunctional CYP genes + massive daily intake of vegetation and water).
2. The Bandhavgarh 2024 die-off was toxicosis from cyclopiazonic acid (CPA) in fungal-infected, unharvested Kodo millet after heavy rains — confirmed by LC-MS.
3. Lesion pattern: acute myocardiopathy → circulatory failure → multi-organ hypoxic/toxic degeneration (liver, kidney, spleen).
4. Kodo millet poisoning is historically documented (since 1933) and has One Health relevance: cattle and humans in the same area are also at risk.
5. Other leading non-infectious causes in India: electrocution (power lines & fences), lightning side-potential (herd deaths), infighting, and nutritional / metabolic deficiencies.
6. Early detection, prompt necropsy, and complete sampling are essential — late-stage decomposition destroys diagnostic evidence.